

Probability Distribution Functions and Data

Complete Detailed Notes for Statistics, Data Analytics, Machine Learning and Interview Revision

This PDF converts the uploaded class notes into an organized, beginner-friendly study booklet with formulas, examples, comparisons and quick revision tables.

What you will learn	Topics covered
Foundation	Random variables, PMF, PDF, CDF, mean, variance and standard deviation
Discrete distributions	Bernoulli trials, Binomial, Poisson and Geometric distributions
Continuous distributions	Uniform, Exponential, Normal and Standard Normal distributions
Data science concepts	Normality tests, z-score, empirical rule and Central Limit Theorem
Interview focus	When to use each distribution, formulas and solved examples

1. Random Variable

A random variable is a variable whose value depends on the result of a random experiment. It converts uncertain outcomes into numbers so that we can apply mathematical and statistical methods.

Example: When a die is rolled, the possible outcomes are 1, 2, 3, 4, 5 and 6. If X represents the number obtained on the die, then X is a random variable.

Types of Random Variables

Type	Meaning	Examples	Function used
Discrete random variable	Takes finite or countable values. Usually whole numbers.	Number of heads, number of calls, number of defective items, number of customers.	PMF - Probability Mass Function
Continuous random variable	Takes any value inside an interval. Values can include decimals.	Height, weight, speed, temperature, waiting time.	PDF - Probability Density Function

Memory trick: Discrete means countable. Continuous means measurable.

2. Probability Distribution

A probability distribution tells how probabilities are assigned to the possible values of a random variable. It answers: what values can X take, and what is the probability of each value?

For example, when rolling a fair die, every value from 1 to 6 has probability $1/6$. The complete set of values with their probabilities forms a probability distribution.

Frequency Distribution vs Probability Distribution

Frequency distribution	Probability distribution
Shows how many times outcomes occurred in observed data.	Shows the probability of possible outcomes.
Based on actual sample counts.	Based on theoretical or estimated probability.
Example: 55 heads in 100 tosses.	Example: probability of head = 0.5.

3. Discrete Probability Distribution and PMF

A discrete probability distribution is a list of possible values of a discrete random variable along with the probability of each value.

PMF Conditions

A Probability Mass Function, or PMF, must satisfy two rules:

1. $p(x) \geq 0$ for every value of x
2. Sum of all $p(x) = 1$

Example: Tossing 3 Coins

Let X be the number of heads when three coins are tossed. The sample space has 8 outcomes: HHH, HHT, HTH, THH, HTT, THT, TTH and TTT.

Number of heads X	0	1	2	3
Probability $P(X)$	1/8	3/8	3/8	1/8

The probabilities add to 1, so this is a valid probability distribution.

4. Continuous Probability Distribution and PDF

A continuous probability distribution describes probabilities for a variable that can take any value in a range. In continuous distributions, the probability at one exact point is zero.

For example, if X is height, $P(X = 170)$ is treated as 0 because there are infinitely many possible decimal values. Instead, we calculate probabilities over intervals like $P(160 \leq X \leq 170)$.

PDF Conditions

$$f(x) \geq 0 \quad \text{Integral from } -\infty \text{ to } \infty \text{ of } f(x) \, dx = 1 \quad P(a \leq X \leq b) = \text{Integral from } a \text{ to } b \text{ of } f(x) \, dx$$

The total area under a PDF curve is always 1. Probability is calculated as area under the curve between two values.

5. CDF - Cumulative Distribution Function

CDF stands for Cumulative Distribution Function. It gives the probability that a random variable is less than or equal to a particular value.

$$F(x) = P(X \leq x)$$

Example: Die Roll CDF

x	1	2	3	4	5	6
$P(X \leq x)$	1/6	2/6	3/6	4/6	5/6	6/6

So, $P(X \leq 3) = 3/6 = 1/2$.

6. Mean, Expected Value, Variance and Standard Deviation

Expected Value / Mean

The mean, also called expected value, is the long-run average value of a random variable when the experiment is repeated many times.

For discrete distribution: $E(X) = \text{Sum of } x * P(X = x)$

Example: Expected Value of a Fair Die

$$E(X) = 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6) \quad E(X) = 21/6 = 3.5$$

This does not mean a die can show 3.5. It means that after many rolls, the average result approaches 3.5.

Variance and Standard Deviation

Variance measures the spread of values around the mean. Standard deviation is the square root of variance and is easier to interpret because it uses the same unit as the variable.

$$\text{Var}(X) = \text{Sum of } p(x) * (x - \text{mean})^2 \quad \text{Alternative formula: } \text{Var}(X) = E(X^2) - [E(X)]^2 \quad \text{SD} = \sqrt{\text{Var}(X)}$$

Solved Example

X	0	1	2	3
P(X)	0.1	0.2	0.4	0.3

$$E(X) = 0(0.1) + 1(0.2) + 2(0.4) + 3(0.3) = 1.9 \quad E(X^2) = 0^2(0.1) + 1^2(0.2) + 2^2(0.4) + 3^2(0.3) = 4.5 \quad \text{Var}(X) = 4.5 - (1.9)^2 = 0.89 \quad \text{SD} = \sqrt{0.89} = 0.9433$$

7. Bernoulli Trials

Bernoulli trials are repeated random experiments where each trial has exactly two outcomes: success or failure.

Conditions for Bernoulli Trials

Condition	Meaning
Fixed number of trials	The number of trials is known in advance.
Independent trials	Result of one trial does not affect another.
Two outcomes	Each trial has success or failure only.
Constant probability	Probability of success remains the same in every trial.

8. Binomial Distribution

Binomial distribution is used when a fixed number of Bernoulli trials are performed and we want the probability of getting a certain number of successes.

When to Use Binomial Distribution

Requirement	Example
Fixed number of trials n	10 bulbs are tested.
Each trial has success/failure	Defective or not defective.
Trials are independent	One bulb result does not affect another.
Probability of success p is constant	Defective probability remains 5%.

Formula

$$P(X = r) = {}^n C_r * p^r * q^{(n-r)} \quad \text{where } q = 1 - p \quad n = \text{total trials} \quad r = \text{number of successes} \quad p = \text{probability of success}$$

Example: Die Rolled 5 Times

Find the probability of getting exactly two ones when a die is rolled 5 times.

$$n = 5, r = 2, p = 1/6, q = 5/6 \quad P(X = 2) = {}^5C_2 * (1/6)^2 * (5/6)^3 \quad P(X = 2) = 10 * (1/36) * (125/216) \quad P(X = 2) = 1250/7776 = 0.1608$$

So the probability is approximately 16.08%.

Mean and Variance of Binomial Distribution

$$\text{Mean} = n p \quad \text{Variance} = n p q \quad \text{Standard deviation} = \sqrt{n p q}$$

Business Example

A digital marketing agency finds that 30% of users who click an online advertisement make a purchase. If 20 users click the ad:

$$n = 20, p = 0.3, q = 0.7 \quad \text{Mean} = np = 20 * 0.3 = 6 \quad \text{Variance} = npq = 20 * 0.3 * 0.7 = 4.2 \quad \text{SD} = \sqrt{4.2} = 2.049$$

9. Poisson Distribution

Poisson distribution is used to model the number of events occurring in a fixed interval of time, area, space or volume.

Examples

Situation	Interval
Calls received at a call center	Per minute or per hour
Patients arriving at emergency room	Per hour
Cars passing a toll booth	Per hour
Printing mistakes in a book	Per page
Accidents on a road	Per day or per month

When to Use Poisson Distribution

Use Poisson distribution when events are counted in a fixed interval, events occur independently, and the average rate of occurrence is known.

Formula

$$P(X = r) = [e^{(-\lambda)} * \lambda^r] / r! \quad \lambda = \text{average number of occurrences in the interval} \quad r = \text{required number of occurrences} \quad e = 2.71828$$

Mean and Variance

$$\text{Mean} = \lambda \quad \text{Variance} = \lambda \quad \text{Standard deviation} = \sqrt{\lambda}$$

Example: Bakery Customers

A bakery receives an average of 5 customers every hour. Find the probability that exactly 3 customers arrive in the next hour.

$$\lambda = 5, r = 3 \quad P(X = 3) = [e^{(-5)} * 5^3] / 3! \quad P(X = 3) = [0.0067 * 125] / 6 \quad P(X = 3) = 0.1404$$

The probability is approximately 14.04%.

Poisson shortcut: If the question says average events per fixed time or space, think Poisson.

10. Geometric Distribution

Geometric distribution models the number of trials needed to get the first success in repeated Bernoulli trials.

Examples

Situation	First success means
Coin tosses	First head
Sales calls	First customer purchase
Basketball shots	First successful basket
Website visits	First conversion

Formula

$P(X = k) = (1 - p)^{(k - 1)} * p$ k = trial number on which first success occurs p = probability of success

Interpretation

Value	Meaning
$P(X = 1) = p$	Success on first trial.
$P(X = 2) = (1-p)p$	Fail first, succeed second.
$P(X = 3) = (1-p)^2 p$	Fail first two trials, succeed third.

Example: First Head on 3rd Toss

$p = 0.5, k = 3$ $P(X = 3) = (1 - 0.5)^{(3 - 1)} * 0.5$ $P(X = 3) = (0.5)^2 * 0.5 = 0.125$

The probability is 12.5%.

Mean and Variance

Mean = $1 / p$ Variance = $(1 - p) / p^2$

11. Uniform Distribution

Uniform distribution is a continuous probability distribution in which every value in an interval is equally likely.

Examples

Selecting a random number between 0 and 1 is uniform. Waiting time for a bus that arrives exactly every 10 minutes can also be modeled as uniform between 0 and 10 if a person arrives randomly.

Formula

$f(x) = 1 / (b - a)$, for $a \leq x \leq b$ Mean = $(a + b) / 2$ Variance = $(b - a)^2 / 12$

Example: Bus Waiting Time

A bus arrives every 10 minutes. A person arrives randomly. Find the probability that waiting time is between 2 and 5 minutes.

$a = 0, b = 10$ Interval length = $5 - 2 = 3$ Total length = $10 - 0 = 10$ $P(2 \leq X \leq 5) = 3/10 = 0.3$

The probability is 30%.

12. Exponential Distribution

Exponential distribution models the waiting time between events that occur continuously and independently at a constant average rate. It is closely related to Poisson distribution.

Poisson vs Exponential

Poisson distribution	Exponential distribution
Counts how many events occur in a fixed interval.	Measures time between two events.
Example: number of calls per hour.	Example: time between two calls.

Formula

$$f(x) = \lambda * e^{(-\lambda x)}, \text{ for } x \geq 0 \text{ Mean} = 1 / \lambda \text{ Variance} = 1 / \lambda^2$$

Memoryless Property

The exponential distribution is memoryless. This means the probability of waiting additional time does not depend on how long you have already waited.

If the question asks about time between events, think Exponential. If it asks about number of events, think Poisson.

13. Normal Distribution

Normal distribution is the most important continuous probability distribution. It is also called Gaussian distribution. It describes data that clusters around the mean in a symmetric bell-shaped curve.

Examples of Normal Distribution

Variable	Typical mean and standard deviation
IQ scores	Mean 100, standard deviation 15
Exam scores	Mean 70, standard deviation 10
Adult height	Mean and standard deviation depend on population
Measurement errors	Often centered around 0

Features of Normal Distribution

Feature	Explanation
Shape	Symmetric bell-shaped curve.
Center	Mean, median and mode are equal.
Area	Total area under the curve is 1.
Mean	Controls the center/location of the curve.
Standard deviation	Controls the spread/width of the curve.
Tails	Extend indefinitely and never touch the x-axis.

Empirical Rule: 68-95-99.7 Rule

Range	Approximate data covered
Mean +/- 1 SD	68%
Mean +/- 2 SD	95%
Mean +/- 3 SD	99.7%

Example

If exam scores are normally distributed with mean 70 and standard deviation 10:

68% of students score between 60 and 80. 95% of students score between 50 and 90. 99.7% of students score between 40 and 100.

14. Normality Tests

Normality tests check whether a dataset can reasonably be considered normally distributed. These are important before applying many statistical tests and machine learning methods.

Shapiro-Wilk Test

Aspect	Explanation
Purpose	Checks whether sample data follows normal distribution.
Outputs	W statistic and p-value.
W close to 1	Data is likely normal.
p-value > 0.05	Accept normality assumption.
p-value <= 0.05	Data is not normal.
Best for	Small to moderate datasets.

D'Agostino-Pearson Test

Aspect	Explanation
Purpose	Checks normality using skewness and kurtosis.
Skewness	Measures asymmetry of data.
Kurtosis	Measures tail and peak shape.
Small statistic and large p-value	Data is close to normal.
Large statistic and small p-value	Data is not normal.
Best for	Large datasets.

Simple Difference

Feature	Shapiro-Wilk	D'Agostino-Pearson
Best for	Small datasets	Large datasets
Based on	Ordered sample values	Skewness and kurtosis
SciPy function	shapiro()	normaltest()

15. Standard Normal Distribution and Z-Score

A standard normal distribution is a normal distribution with mean 0 and standard deviation 1. It is represented by Z.

Mean = 0 Standard deviation = 1

Z-Score Formula

$Z = (X - \text{mean}) / \text{standard deviation}$

A z-score tells how many standard deviations a value is away from the mean.

Z-score	Meaning
0	Exactly at the mean.
1	One standard deviation above the mean.
-1	One standard deviation below the mean.
2	Two standard deviations above the mean.

Example

Exam marks are normally distributed with mean 70 and standard deviation 10. A student scores 85.

$Z = (85 - 70) / 10 = 1.5$

This means the student scored 1.5 standard deviations above the mean.

16. Central Limit Theorem

The Central Limit Theorem, or CLT, explains why normal distribution appears so often in real-world data analysis.

It states that if you take many random samples from any population and calculate the sample means, the distribution of those sample means becomes approximately normal as sample size increases, even if the original population is not normal.

Mathematical Idea

If $X_1, X_2, \dots, X_n$ are independent and identically distributed, then sample mean $\bar{X}$ approximately follows: $\bar{X} \sim \text{Normal}(\text{mean}, \text{variance} / n)$ Standard error = $\sigma / \sqrt{n}$

Why CLT is Important

Reason	Explanation
Works with non-normal data	Sample means become approximately normal for large samples.
Foundation of inference	Used in confidence intervals and hypothesis testing.
Useful in analytics	Helps estimate population means using samples.
Practical importance	Allows normal probability models in many real-world cases.

17. Distribution Selection Guide

Question pattern	Use this distribution
Fixed number of trials and success/failure outcome	Binomial
Number of events in fixed time or space	Poisson
Number of attempts until first success	Geometric

Every value in a range is equally likely	Uniform
Waiting time between events	Exponential
Bell-shaped natural data around mean	Normal
Convert any normal value to a universal scale	Standard normal / Z-score

18. Complete Formula Sheet

Concept	Formula
PMF condition	$p(x) \geq 0$ and $\sum p(x) = 1$
PDF condition	$f(x) \geq 0$ and total area under curve = 1
CDF	$F(x) = P(X \leq x)$
Expected value	$E(X) = \sum x P(X = x)$
Variance	$\text{Var}(X) = E(X^2) - [E(X)]^2$
Standard deviation	$SD = \sqrt{\text{Var}(X)}$
Binomial	$P(X = r) = {}^nC_r p^r q^{(n-r)}$
Binomial mean and variance	Mean = np , Variance = npq
Poisson	$P(X = r) = e^{(-\lambda)} \lambda^r / r!$
Poisson mean and variance	Mean = λ , Variance = λ
Geometric	$P(X = k) = (1-p)^{(k-1)} p$
Geometric mean and variance	Mean = $1/p$, Variance = $(1-p)/p^2$
Uniform PDF	$f(x) = 1/(b-a)$
Uniform mean and variance	Mean = $(a+b)/2$, Variance = $(b-a)^2/12$
Exponential PDF	$f(x) = \lambda e^{(-\lambda x)}$
Exponential mean and variance	Mean = $1/\lambda$, Variance = $1/\lambda^2$
Z-score	$Z = (X - \text{mean}) / \text{standard deviation}$
Standard error in CLT	$\sigma / \sqrt{n}$

19. Interview-Level Quick Revision

Question	Answer
What is a random variable?	A variable whose value depends on the outcome of a random experiment.
Difference between PMF and PDF?	PMF is for discrete variables; PDF is for continuous variables.
Why is $P(X = x)$ zero for continuous variables?	Because a continuous variable has infinitely many possible exact values; probability is measured over intervals.
When do we use binomial distribution?	When there are fixed independent trials with two outcomes and constant probability of success.
When do we use Poisson distribution?	When counting events in a fixed interval with known average rate.
When do we use geometric distribution?	When counting trials until the first success.
What is λ in Poisson?	The average number of events in the fixed interval.

What is the standard normal distribution?	A normal distribution with mean 0 and standard deviation 1.
What does z-score tell?	How many standard deviations a value is from the mean.
What is CLT?	Sample means become approximately normally distributed as sample size increases.

20. Final Summary

Probability distributions help us understand uncertainty. They describe how likely different outcomes are and help us calculate probabilities, expectations, variance and rare events.

For countable data, use discrete distributions such as Binomial, Poisson and Geometric. For measurable data, use continuous distributions such as Uniform, Exponential and Normal.

In data science and analytics, these concepts are used for statistical testing, probability modeling, predictive analytics, anomaly detection, machine learning assumptions and business decision making.

Final memory trick: Binomial = fixed trials. Poisson = count events. Geometric = first success. Uniform = equal chance. Exponential = waiting time. Normal = bell curve.